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PAPER

Size- and shape-controlled synthesis of Prussian Blue nanoparticles by a polyvinylpyrrolidone-assisted crystallization process

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The preparation of coordination polymer nanoparticles is currently the focus of research attention because of the advantage of increasing the specific surface area. In this paper, we report a facile method to prepare Prussian Blue (PB) nanoparticles with different particle sizes. Their particle sizes have a wide range from a few dozen nanometres to a submicrometre, depending on the synthetic conditions. The obtained particles can be divided into three categories: small (around 20 nm), medium-sized (around 100 nm), and large (in excess of 200 nm). The medium-sized and large particles are formed by oriented aggregation of fine nanoparticles, showing a mesocrystal conformation (*i.e.*, pseudo-single crystal nature). Our results strongly suggested that non-classical crystallization dominates the present system, which can help to further control and understand the crystallization process of other coordination polymers. By means of N₂ gas adsorption–desorption measurements, the nanoporosity and specific surface areas for typical samples with three different particle sizes are characterized. Among these samples, medium-sized PB particles show the highest surface area of 260 m² g^{−1}, while large PB particles exhibit very poor porosity. The results show that the gas uptakes of coordination polymers can be adjusted by the particle size and shape.

1. Introduction

Coordination polymers or metal–organic frameworks, as one-kind of human-designed materials, show flexible structural and functional tenability.^{1,2} They have been applied in various fields, including adsorption, sensing, catalysis, separation, and gas storage.² To date, the main research regarding coordination polymers has focused on bulk scale crystals to easily characterize their microstructures and basic properties. Nano-sized particles offer great opportunities for tuning the properties and improving the performance of materials, as has been demonstrated with many materials. Inspired by such cases, it is of great interest to explore nano-sized coordination polymers.^{3,4} During a pioneering study, nano-sized coordination polymers showed distinct properties compared to bulk crystals.^{3,4a,b,d} For example, when coordination polymers were made into nanoparticles or thin films,^{4a,b,d,5} improved sensitivity and accelerated adsorption capacity were observed. The outstanding results have

encouraged others in the field of material science to study nano-sized coordination polymers.

Although the size- and shape-controlled synthesis of coordination polymers has been underway for a few years, some progress has already been made by applying limited spaces. Mesoporous silica,⁶ emulsion droplets,^{3b,g,7} and microfluid³ⁱ could reduce the particle size of coordination polymers. However, these templating methods are difficult with regard to tuning the size of particles in a wide range or satisfactorily controlling the morphology of products. To overcome such disadvantages, several template-free methods (*e.g.*, surfactant-assisted and rapid heating processes) were developed.^{3e,f,h,4,8} Various shapes, such as nanoparticles, nanorods, nanowires, and polyhedrons, have been reported. A few reports have shown good tunability for particle sizes and shapes, but no systematic study has been reported because of the short research history. Some questions remain, for example, (a) whether there is method to be used to control the size of coordination polymers in a wide range, (b) what the driving force for the morphological change is, and (c) whether the crystallization mechanism of coordination polymers is similar to that of other materials. Therefore, it is of fundamental importance to perform a systematic research on the preparation of coordination polymers from nanometre to micrometre scales.

Prussian Blue (PB), as an important coordination polymer, has received considerable attention in the past decades because of its wide applications in various fields.⁹ Inspired by size-dependent properties,^{3f,8a,c} the controlled synthesis of PB is an important aspect in coordination polymer science. To date, PB

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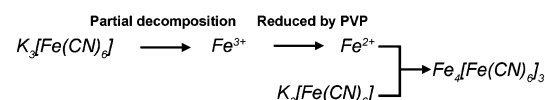


Fig. 1 Reaction scheme for the preparation of PB particles.

particles with different sizes and morphologies have been synthesized by several methods.^{6,7a-c,8a-d,10} However, a facile method that can be used to prepare PB particles with controlled sizes (from nanometre to micrometre scale) is still lacking. In addition, the crystallization mechanism of PB particles is still unclear. Therefore, it would be very interesting to determine whether classical or non-classical crystallization dominates the crystallization of PB.¹¹

In this paper, we demonstrate the size- and shape-controlled syntheses of PB nanoparticles with a polyvinylpyrrolidone-assisted crystallization process. This method provides for the easy control of PB particle sizes. The obtained particles can be classified into three types: small (around 20 nm), medium-sized (around 100 nm), and large (in excess of 200 nm). Small PB and medium-sized PB particles had higher surface areas than the large PB particles. The whole reaction was based on the partial decomposition and reduction of $K_3[Fe(CN)_6]$ in acidic solution with the addition of polyvinylpyrrolidone (PVP), as shown in Fig. 1. To investigate the role of several reaction parameters on the size and shape of the final particles, we investigated the effects of the $K_3[Fe(CN)_6]$ concentrations, the PVP amounts, and the pH values of the initial solutions. In addition, we checked the N_2 gas adsorption–desorption reactions of different products. These products had interesting size-dependent properties. This work is of fundamental importance in the controlled synthesis of PB particles. Furthermore, it will contribute to an understanding of the crystallization mechanism of other coordination polymers with novel properties in future.

2. Experimental

2.1. Synthesis

The reaction scheme for the preparation of PB is shown in Fig. 1. In a typical procedure, PVP (K30) and $K_3[Fe(CN)_6] \cdot 3H_2O$ were added to a 40 mL HCl solution under magnetic stirring. After 30 min of stirring, a clear solution was obtained. The added PVP amounts varied from 1.0 g to 10 g, and the $K_3[Fe(CN)_6]$ concentrations were in the range from 0.01 to 0.05 mol L⁻¹. The pH values of the solutions varied from 0 to 1 and 2. The solution was then sealed in a vial, placed into an electric oven, and heated at 80 °C for 20 h. After aging, the blue precipitates were collected by centrifugation and washed in distilled water and ethanol several times. After drying in a vacuum oven at room temperature for 12 h, PB powder was obtained. The obtained particles can be classified into three types: small (around 20 nm), medium-sized (around 100 nm), and large (in excess of 200 nm).

2.2. Characterization

SEM images were taken with a Hitachi S-4800 scanning microscope at an accelerating voltage of 10 kV. A TEM observation was performed using a JEM-2010 TEM system that was operated at 200 kV. Wide-angle powder X-ray diffraction (XRD) patterns

were obtained with a Rigaku RINT 2500X diffractometer using monochromated Cu K α radiation (40 kV, 40 mA) at a scanning rate of 1° min⁻¹. Nitrogen adsorption–desorption analyses were performed using a Quantachrome Autosorb Automated Gas Sorption System at 77 K. Before the measurement, the PB particles were preheated at 40 °C under a dynamic vacuum condition for 60 h to remove the adsorbed water molecules. Real-time monitoring of Cs ion adsorption was recorded by quartz crystal microbalance (QCM) measurement (QCA922, SEIKO EG&G Co., Japan). Before the PB coating, a QCM electrode was sonicated in a mixture of ethanol and water. After being rinsed with copious amounts of water and dried with a N₂ flow, the frequency was recorded as the initial frequency (f_0) for the QCM electrode. The PB films coated on the electrode of QCM were prepared as follows. The PB particles (1 mg mL⁻¹) were dissolved in an aqueous solution of Nafion (0.1 wt%) and then the mixtures were sonicated for 30 min. The mixtures were dropped onto the QCM electrodes. After drying under gentle nitrogen flow, the electrode surfaces were washed with pure water and then dried overnight under vacuum. Firstly, we measured the real-time monitoring of electrode for water (50 mL) until the steady state at which no more water molecules adsorbed into the PB films. This frequency shift corresponds to the adsorption of water molecules into the PB film. After that, the electrode was dried up under vacuum and the second real-time monitoring was conducted by immersing the electrode in 1 ppm Cs solution (50 mL). This frequency shift is due to the adsorption of Cs solution (Cs ions + water molecules) into the PB film. Therefore, by subtracting the water adsorption part from the second monitoring data, the frequency shift, Δf (Hz), corresponding to only the amount of Cs ions, can be calculated. All experimental measurements were carried out in an air-conditioned room at 24 °C.

QCM is sensitive to mass change, when the surface of a quartz crystal electrode is coated with a sensitive coating material. Its gravimetric sensing ability allows quantitative analysis of molecular adsorption on the nano-gram scale and real-time monitoring of sorption processes. The mass per unit area of PB films, m' (g cm⁻²), deposited onto the gold electrodes of QCM, can be measured by the oscillating frequency of the quartz electrodes. The relation between the frequency shifts, Δf (Hz), corresponds to the sample amounts loaded on the QCM electrode and m' is well-established from the work of Sauerbrey.¹²

$$\Delta f = - (2f_0^2 / \sqrt{\rho_0 \mu_0}) m'$$

where f_0 (Hz) is the natural frequency of the quartz crystal, ρ_Q is the quartz density (2.649 g cm^{-3}), and μ_Q is the shear modulus ($2.947 \times 10^{11} \text{ g cm}^{-1} \text{ s}^{-2}$).

The small amount of PB particles deposited on QCM electrode ($24 \mu\text{g cm}^{-2}$) presented in the form of a thin film, additionally result in a fast signal response due to the short diffusion times. Thus, the diffusion of Cs ions into PB particles were studied in terms of diffusion coefficient (D) using a simple linear diffusion model of Fick's second equation, which has been reviewed by Crank.¹³

$$\frac{\Delta f_t}{\Delta f_\infty} = 4\sqrt{\frac{D}{\pi}} \frac{t^{1/2}}{L}$$

where Δf_t is the frequency change due to the diffusion of the Cs ions into PB particles at any time t and Δf_∞ is the frequency change in the equilibrium state at the end of the adsorption process. These two parameters can be given by $\Delta f_t = f - f_t$ and $\Delta f_\infty = f - f_\infty$ (f is the frequency of the oscillator before the immersion), respectively.

3. Results and discussion

3.1. Crystal growth of PB particles with different sizes

We first fixed the initial pH value of solution at 0 and changed the $K_3[Fe(CN)_6]$ concentrations and PVP amounts from 0.01 to 0.05 mol L⁻¹ and 1 to 10 g, respectively. SEM images of all the products are shown in Fig. 2. Two types of PB particles were observed: small PB nanoparticles (around 20 nm, Fig. 2b, c, and f) and medium-sized PB particles (around 100 nm). To further analyze the microstructure, the small PB nanoparticles were observed by TEM (Fig. 3a and b). Most of the nanoparticles were isolated from each other, but some particles were aggregated on the TEM grid. From a highly magnified TEM image, one particle possessed several domains with different crystal orientations, although the lattice fringes were clearly observed over the entire area of the particle (Fig. 3a-2). The other kind of

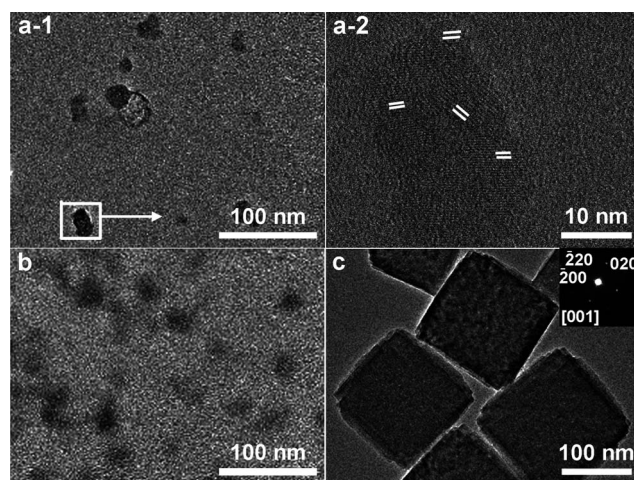


Fig. 3 TEM images of PB particles. (a-1) Low-magnified TEM image of small PB particles (see Fig. 2c for the SEM image). (a-2) High-resolution TEM image taken from a single particle shown with the white frame in (a-1). (b) Low-magnified TEM image of small nanoparticles (see Fig. 2f for the SEM image). (c) Low-magnified TEM image of medium-sized particles (see Fig. 2j for the SEM image). The inset is the selected area ED pattern taken from a single particle.

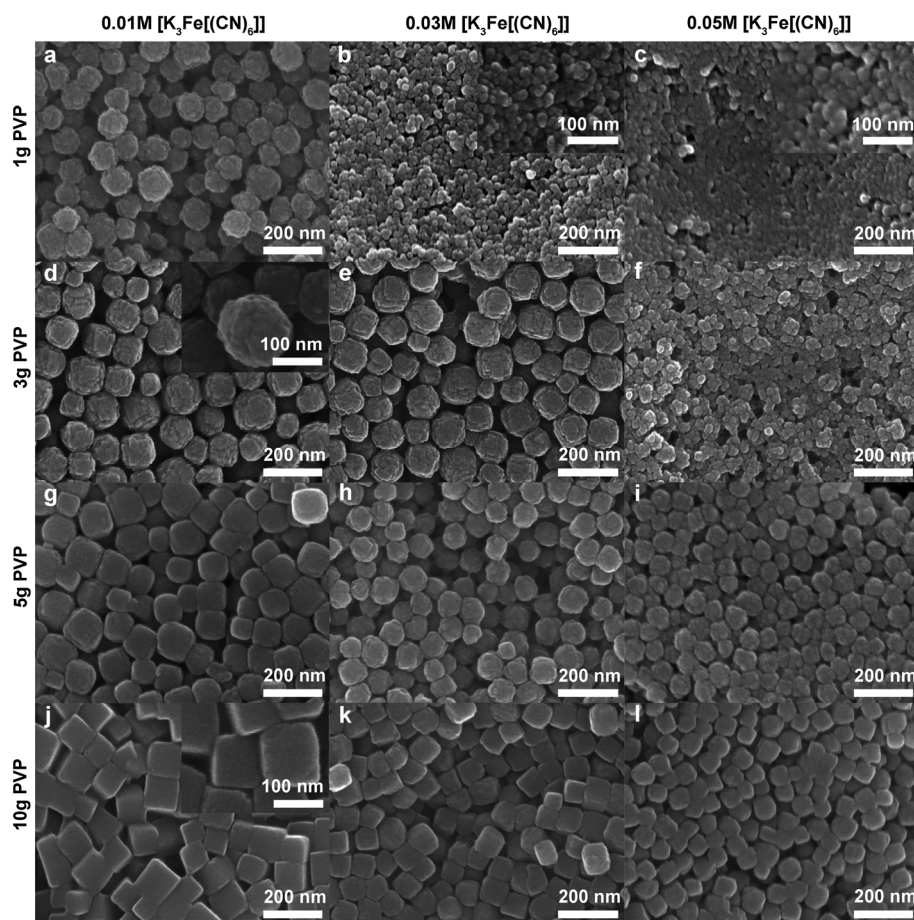


Fig. 2 SEM images of PB particles prepared when the pH value of solutions was fixed to be 0. Insets (b), (c), (d), and (j) represent the highly magnified images of these samples.

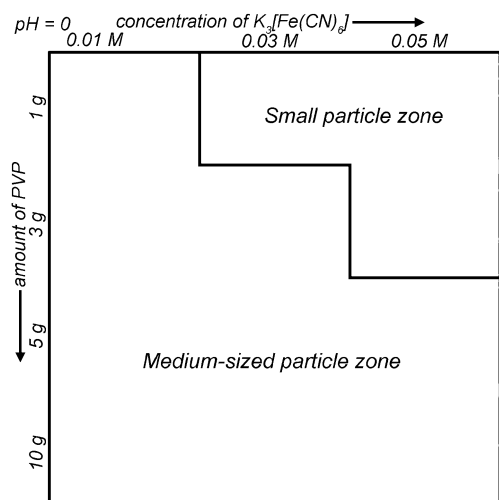


Fig. 4 Particle-type classification of the PB products prepared when the pH value of solutions was fixed at 0.

product (medium-sized PB particles) showed a uniform particle-size distribution (average size, around 100 nm). Enlarged SEM images taken from selected particles (the inset of Fig. 2d and j) revealed that the surfaces or the edges of these particles, where the very fine PB nanoparticles would be aggregated, were not smooth. (The details are given in a later section.) Moreover, selected-area electron diffraction (ED) patterns taken from one particle presented single-crystal like diffraction spots (Fig. 3c), suggesting an oriented arrangement of the fine PB nanoparticles.

On the basis of the above SEM results, particle-type classification of the PB products prepared when the pH value of solutions was fixed to be 0 is shown in Fig. 4. The formation of the small PB nanoparticles was observed in the area with a low PVP amount and high $K_3[Fe(CN)_6]$ concentration, while the formation of the medium-sized PB particles took place in other areas in the diagram. With the increase of the PVP amounts and decrease of the $K_3[Fe(CN)_6]$ concentration, the medium-sized particles gradually changed to well-defined cubes. In addition, as clearly seen in Fig. 2, the increase of $K_3[Fe(CN)_6]$ concentration decreased the particle sizes (see Fig. 2 from the left side to the right side).

The pH value was increased to 1, while the $K_3[Fe(CN)_6]$ concentrations and PVP amounts were changed from 0.01 to 0.05 mol L⁻¹ and 1 to 10 g, respectively. Fig. 5 illustrates the SEM images of the obtained products. On the basis of the SEM images, particle-type classification is also shown in Fig. 6. The small PB nanoparticle region, which is evident in Fig. 4, decreased in size. The products shown in Fig. 5b contained both small and medium-sized particles. As evident in Fig. 5k and l, large PB particles (larger than 200 nm) are present. These particles with smooth surfaces are sometimes rectangular. The formation mechanism of the large PB particles is explained in a later section. The results above suggest that both the PVP amounts and pH values are a key factor for the generation of the large particles.

After increasing the pH value to 2, no small PB nanoparticles were observed (Fig. 7). Compared to those in Fig. 4 and 6, large particle regions are observed in Fig. 8. Obviously, increasing the

pH (from 0 to 1 and 2) favored the generation of large particles and also strictly hindered the formation of small nanoparticles. In addition to the pH values, the high concentration of $K_3[Fe(CN)_6]$ could also prompt the formation of large particles. For instance, when the PVP amount was kept at 3.0 g, the particle sizes increased with the increase of the $K_3[Fe(CN)_6]$ concentrations (Fig. 7d–f).

Particles of varying size in three categories were obtained by systematically tuning the reaction parameters (Fig. 4, 6, and 8). The phase purity of these products (Fig. 2c and e, and 7l) was examined by wide-angle XRD (Fig. 9). The XRD patterns of three typical products were almost identical, which also coincided with the standard patterns of the PB crystal (JCPDS card, 73-0689). Therefore, all the products were exactly pure PB, although the peak intensities and widths varied with the particle sizes. For small PB nanoparticles, broader peaks with low intensity were observed due to the small particle size (around 20 nm) and poor crystallinity shown in the TEM observation (Fig. 3a). The diffraction peaks for medium-sized particles were sharper, and FWHMs (full width at half maximums) were narrower. With a further increase in the particle sizes, the observed XRD peaks became very sharp and strong, compared to others, due to the formation of large particles with a pseudo-single crystal nature.

To understand the formation mechanism, the crystallization process was carefully investigated by a TEM study. To investigate the generation process of the medium-sized PB particles (as shown in Fig. 2e) and large PB particles (as shown in Fig. 7l), samples formed at the early growth stage were taken (Fig. 10). At the early growth stage of the medium-sized particle, quasi-cubic particles were observed (Fig. 10a). The presence of porous structure (indicated by arrows) indicated that quasi-cubes were formed by the aggregation of very small nanoparticles (diameters around 5 nm). Selected-area ED patterns taken from one particle showed several intense spots assignable to PB single crystal, indicating that the medium-sized particles had been formed by aggregation of small nanoparticles in a highly ordered way. Similar to the growth of medium-sized particles, at the early growth stage of the large particles, ordered aggregation of nanoparticles was also confirmed, and quasi-cube-shaped particles were mainly observed (Fig. 10b). After one hour, rectangular particles with very sharp corners were observed, and some of them looked like assemblies of two preformed cubes. Selected-area ED patterns taken from the whole particles gave single-crystal-like periodic patterns, which strongly suggested that fine nanoparticles were assembled in an oriented way. Therefore, on the basis of the observation, the generation of large particles included the formation of medium-sized particles and their subsequent assembly (Fig. 11).

It is well known that general crystallization involves nucleation and growth processes. If the concentration of the solute is much higher than the critical nucleation point, explosive nucleation could take place, providing large numbers of nuclei and generating small particles.¹⁴ When sufficient additives, such as soluble polymers or surfactants, exist, these additives could coat the surfaces of preformed nanoparticles, then reduce the surface tension of particles, and temporarily stabilize them in solution. However, the stabilization cannot be constantly maintained because of the remaining high surface energy; therefore, these

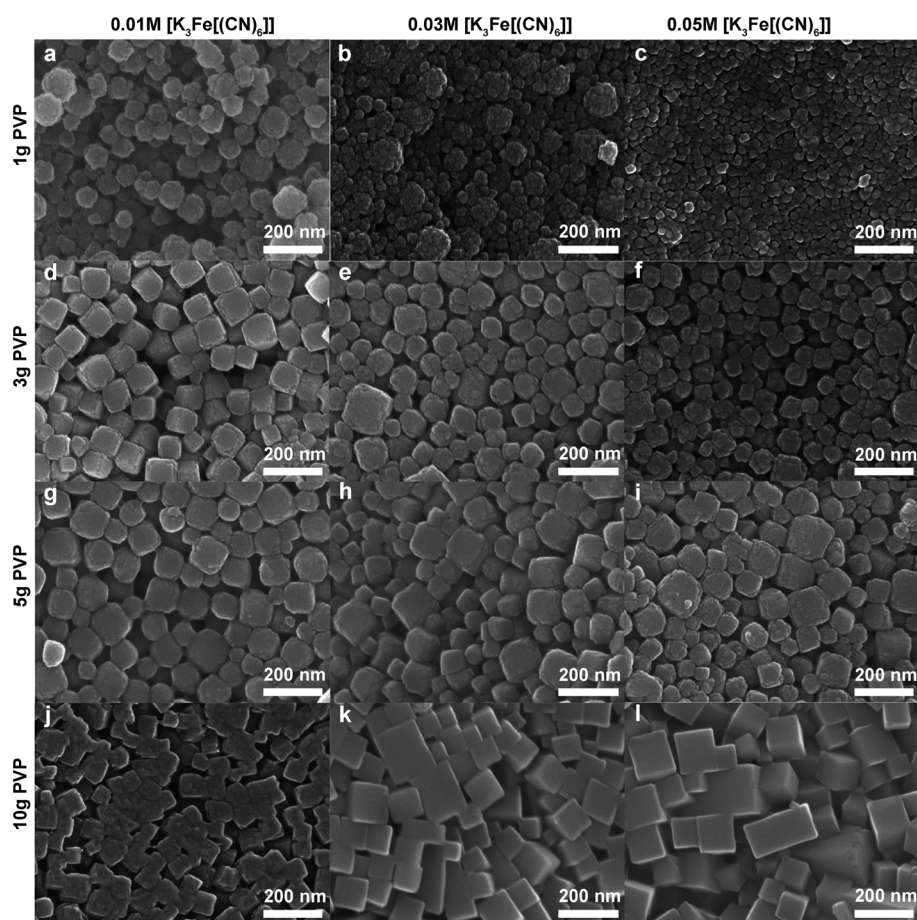


Fig. 5 SEM images of the PB particles prepared when the pH value of solutions was fixed at 1.

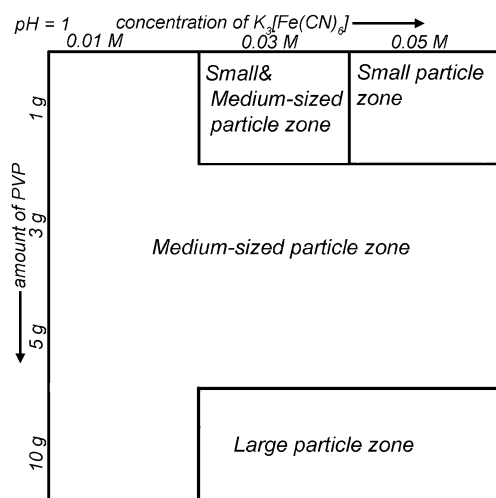


Fig. 6 Particle-type classification of the PB products prepared when the pH value of solutions was fixed at 1.

nanoparticles tend to aggregate in an ordered way to form a superstructure (Fig. 11).¹⁵ If the superstructure shows a single-crystal-like ED pattern and rough surfaces or porous structures, they are referred to as mesocrystals.¹⁵ Sometimes, mesocrystals are stable, in which case they are final products. However, sometimes, mesocrystals are just intermediates; they could

further convert into real single crystals by self-driving fusion.¹⁵ At this time, the rough surfaces become smooth, and the porous structures transform into dense structures. In our case, a multiple-step mechanism could be proposed to explain the growth of PB particles, as shown in Fig. 11. At the initial stage, explosive nucleation was preferred during the formation of PB, leading very easily to a small nanocrystal, which we refer to as a small nanoparticle. When there were sufficient amounts of PVP, the surface of the small particles could be coated. Thus, PVP could temporarily stabilize them in solution at first. Owing to the tendency to reduce the total surface energy, the small particles aggregated in an ordered way to form mesocrystals, which were medium-sized particles, to reduce the surface energy. If the medium-sized particles were not stable because of the remaining high surface energies caused by internal defects and pores, they would further aggregate to reduce the surfaces. In the meantime, they could also transform into dense structures to erase pores. After these transformations, medium-sized particles finally became large. Such a multiple-step growth process is a typical non-classical crystallization, which is observed commonly in biomineralization.¹⁵ During these non-classical crystallizations, the existence of polymers plays critical roles in most cases. In our synthesis, we used PVP as a soluble polymer.

On the basis of the mechanism above, we understand how the size and morphology of PB particles were determined by the reaction parameters, such as the concentration of $K_3[Fe(CN)_6]$,

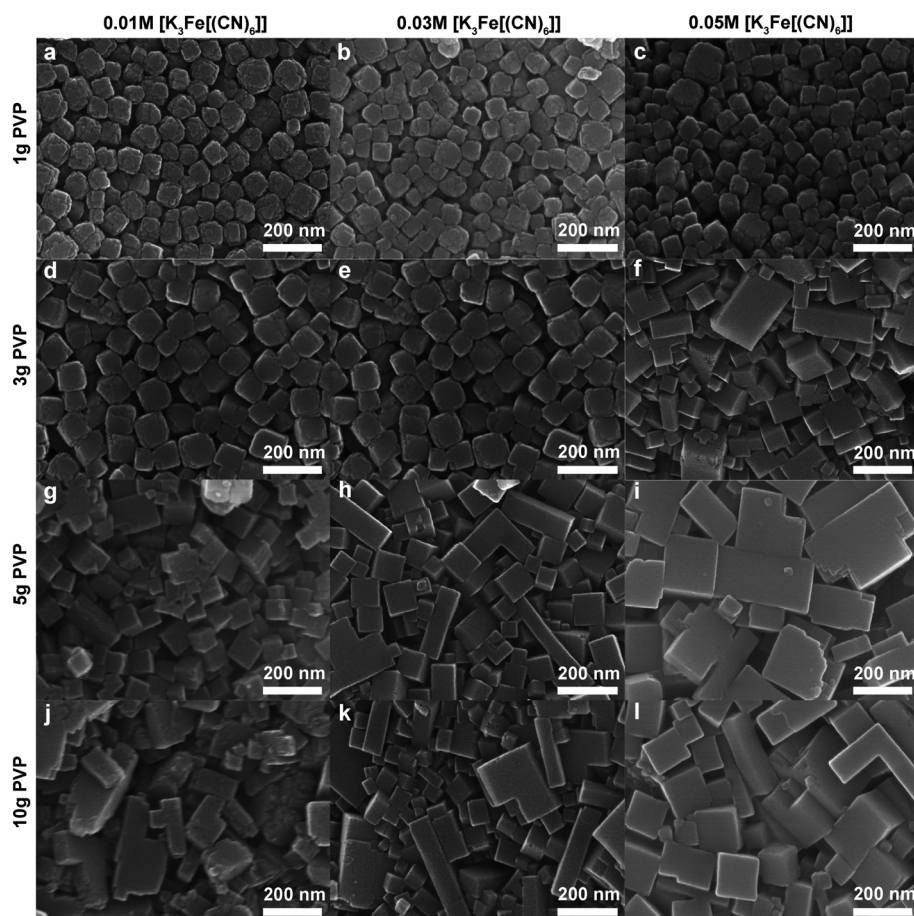


Fig. 7 SEM images of the PB particles prepared when the pH value of solutions was fixed at 2.

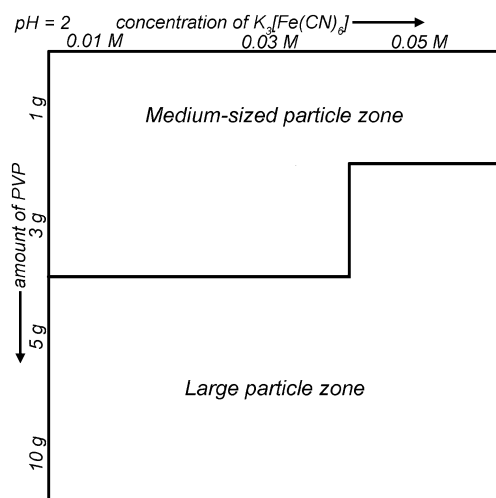


Fig. 8 Particle-type classification of the PB products prepared when the pH value of solutions was fixed at 2.

the amounts of PVP, and the pH values of solution. The concentration of $K_3[Fe(CN)_6]$ represented the source of PB crystals. A higher concentration could elevate the nucleation rate and provide more nuclei. The large number of nuclei means that their growth consumes more resources, making it difficult for the particles to grow larger. In addition, the large number of particles

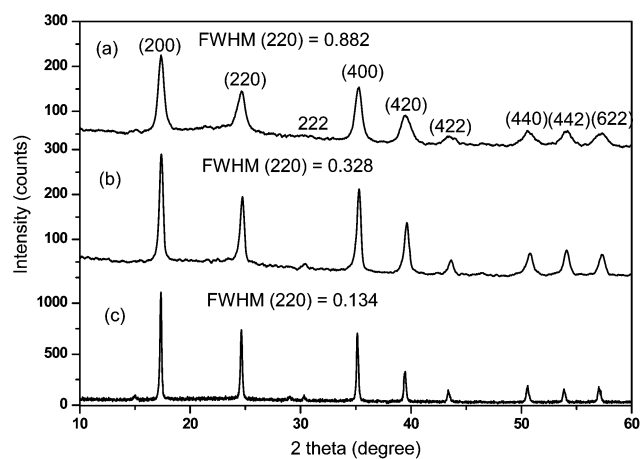


Fig. 9 Wide-angle XRD patterns of different types of PB particles ((a) small PB nanoparticles shown in Fig. 2c, (b) medium-sized PB particles shown in Fig. 2e, and (c) large PB particles shown in Fig. 7l).

indicates that the total surface energy was high, which would make the system unstable. When a system was not stable, the particles would aggregate to reduce the surface energy. As a result, a high concentration of $K_3[Fe(CN)_6]$ could cause the products to stay in the small-sized nanoparticle zone (when PVP could not stabilize them) or become large by aggregation of

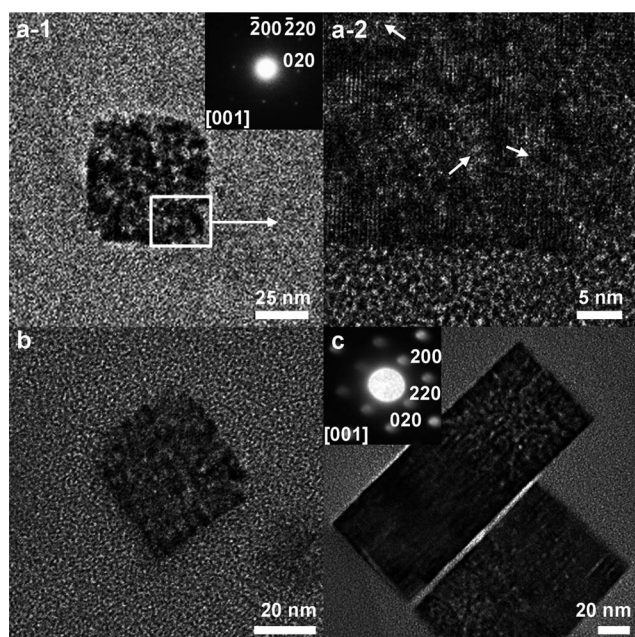


Fig. 10 (a) TEM image and SAED pattern of a medium-sized PB particle grown after 30 min. Reaction conditions: pH = 0, PVP = 3 g, and $[K_3[Fe(CN)_6]] = 0.03$. (a-2) High-resolution TEM image of a partial particle shown by a white rectangular frame in (a-1). The white arrows in (a-2) illustrate the pores. (b) and (c) TEM and ED patterns of a large PB particle grown after 30 min and 1 h, respectively. Reaction conditions: pH = 2, PVP = 10 g, and $[K_3[Fe(CN)_6]] = 0.05$.

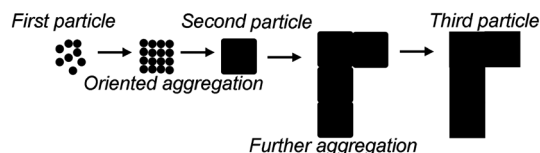


Fig. 11 Growth process of large PB particles.

medium-sized particles (when the nucleus could be stabilized by PVP), while lower concentration of $K_3[Fe(CN)_6]$ could cause the products to stay in the medium-sized particle zone. PVP used in our system mainly served as a capping agent that could stabilize the nucleus at first. If the amount of PVP was insufficient, particles would directly precipitate without further growth. If the amount of PVP was sufficient, the nucleus would first stabilize and then self-assemble in an ordered way. Therefore, a small amount of PVP favored the products in the small nanoparticle zone, while a large amount of PVP favored the products in the medium-sized and large particle zone. The pH values of the solution represented the concentration of hydrochloride, which determined the decomposition rate of $K_3[Fe(CN)_6]$; therefore, they controlled the generation rate of PB crystals. A higher pH value accelerated the generation rate of PB crystals, which meant that a larger number of PB nuclei formed simultaneously, which facilitated the generation of small nanoparticles, while a lower pH value indicated that a small number of PB nuclei, which could be easily coated by PVP, formed at first, resulting in the formation of medium-sized and large particles. The discussion above suggests that the roles of the concentration of

$K_3[Fe(CN)_6]$, the amounts of PVP, and the pH values of the solution strongly interacted with each other; therefore the realistic formation mechanism of PB crystals was complex, resulting in the formation of products of various sizes and morphologies.

3.2. Adsorption properties of PB particles with different sizes

It is generally known that PB is a microporous material when the water molecules are removed.^{9,16} Owing to the highly porous structure, PBs and their analogues show great potential in gas adsorption. However, the size/shape-dependent adsorption property of PB has not been explored yet. Therefore, we selected three kinds of PB particles with different particle sizes (as shown in Fig. 2c and e and 7l) and degassed them at 40 °C under a dynamic vacuum condition for 60 h. The related N_2 gas adsorption–desorption isotherms are shown in Fig. 12. At low $P/P_0 = 0–0.1$ (microporous region), the N_2 uptakes for small and medium-sized nanoparticles were much larger than those for the large particles. For small and medium-sized particles, at higher $P/P_0 = 0.3–0.9$, the adsorbed amount of N_2 gas increased gradually, and significant hysteresis loops were observed. The BET surface areas were 206 $m^2 g^{-1}$ (for small PB nanoparticles) and 254 $m^2 g^{-1}$ (for medium-sized PB particles). The pore-size distribution curves measured by the Barrett–Joyner–Halenda (BJH) method showed the existence of mesopores that were created by a void among the assembled PB particles. In contrast, large particles exhibited very small N_2 uptakes, and the hysteresis loop was not obvious. The existence of mesopores was not confirmed. The surface area exhibited a limited value of only about 60 $m^2 g^{-1}$. Thus, the differences among the N_2 gas adsorption of the PB particles came from the varying particle sizes.

In previous reports, the size of porous coordination polymers (PCPs) could influence the N_2 gas adsorption.^{4a–c} However, the differences generally occurred at $P/P_0 = 0.3–0.9$; therefore, the BET surface areas of PCP nanoparticles were slightly larger than those of bulk crystals due to the existence of mesopores.^{4a–c} However, in our case, the large particles showed very poor porosity compared to smaller particles. Large PB particles and their dense crystal structure would be obstacles for removing water molecules during a degassing treatment, thereby inhibiting the generation of micropores and leading to low surface areas. For small and medium-sized PB particles, the short diffusion distance to the outside can easily allow the water molecules to move; thus, the microporosity could be successfully generated after degassing, leading to a high surface area. Medium-sized PB particles realized higher surface areas, compared to small PB nanoparticles. The reason may be that the crystallinity of medium-sized PB particles was better than that of the small PB nanoparticles. The medium-sized PB particles showed a single crystalline nature (Fig. 3c), while the small PB nanoparticle contained several crystal domains (Fig. 3a-2). A coherent crystalline structure was able to generate well-developed microporous structures after degassing, resulting in a higher surface area. According to the previous papers, amorphous or poorly crystalline MOFs showed poor porosity and very low specific surface area, compared to highly crystalline MOFs.¹⁷ Therefore, highly crystallized frameworks are necessary to realize high surface areas.

Based on our PB materials, we set out to study how effectively PB particles were able to adsorb Cs^+ ions from aqueous solution.

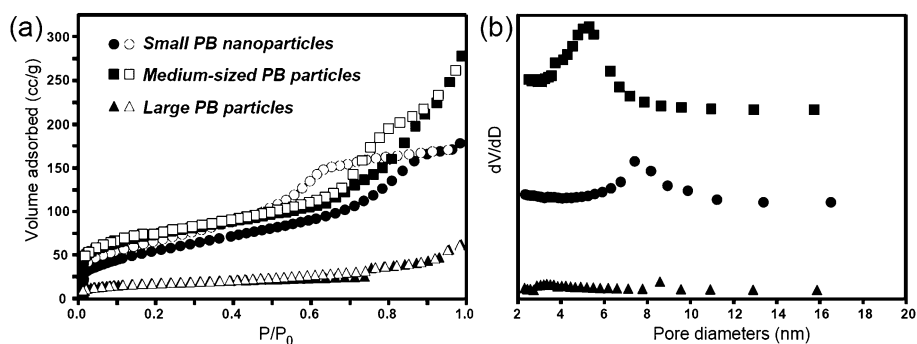


Fig. 12 (a) N₂ adsorption-desorption isotherms of different types of PB particles: (●) small PB nanoparticles, as shown in Fig. 2c, (■) medium-sized PB particles, as shown in Fig. 2e, and (▲) large PB particles, as shown in Fig. 7l. (b) Size distribution of mesopores of different types of PB particles calculated by the BJH method.

It has been well known that PB is used for environmental cleanup of Cs⁺ ions and also is approved by the FDA for chelation therapy of Cs. Advancement of nuclear technology has resulted in large-scale release of radioactive Cs isotopes into environment. Since ¹³⁷Cs is one of the main fission products in radioactive wastes, the release of ¹³⁷Cs into environment damages the ecosystem owing to its long half-life and unlimited solubility, which causes its migration to the biosphere through ground water and tends to remain in an available form to biota for several decades.¹⁸ On the other hand, however, the radioactive Cs is an excellent source for some clinical and biotechnological applications like surgical instrument disinfection, radiotherapy, and food sterilization. Therefore, effective collection of the radioactive Cs from waste solutions is currently a significant issue.¹⁹

Here, Cs adsorption behavior was studied by using the QCM technique. In order to investigate such adsorption capability, a fast and sensitive sensor based on PB particles coating on a QCM electrode was prepared with the aid of a Nafion binder. In that case, small amounts of the PB particles (24 μg cm⁻²) deposited on the gold electrode of QCM were immersed in a solution of 1 ppm Cs⁺ ions. The frequency of the quartz crystal electrode was immediately changed and decreased due to the fast adsorption uptake of Cs⁺ ions into the PB films. The time-dependence of frequency shift (Δf), which corresponds to the

amounts of Cs⁺ ions adsorbed onto the PB particles, is shown in Fig. 13a. We clearly observed a large adsorption uptake for Cs⁺ ions into the medium-sized PB particles ($\Delta f = 200$ Hz). Medium-sized PB particles accommodated more than 2 times higher uptakes for Cs⁺ ions than that for large PB particles. Additionally, the loading capacities of small-, medium- and large-sized PB particles for Cs⁺ ions were 53 mg g⁻¹, 68 mg g⁻¹, and 25 mg g⁻¹, respectively. These data coincided with those obtained by ICP analysis (58 mg g⁻¹ for small-sized PB particles, 78 mg g⁻¹ for medium-sized PB particles, and 35 mg g⁻¹ for large-sized PB particles). The large different adsorption capability between them was attributed to the specific surface areas. It has been generally accepted that PB can be divided into two structural categories; water-soluble and water-insoluble PB.²⁰ For the water-soluble PB, K⁺ ions are included in the crystal structure. But, for the water-insoluble PB, K⁺ ions are generally excluded. In our case, we prepared the water-insoluble PB particles which had no K⁺ ions in the crystal structure. Therefore, the Cs adsorption was caused by insertion into the microporosity, rather than exchange of K ions.

More interestingly, the particle sizes of PB significantly affected the rate of adsorption uptake, which is controlled by diffusion. The diffusion in terms of diffusion coefficient (D) was investigated by real-time monitoring of the frequency shift after the adsorption of Cs ions onto the PB particles. The experimental

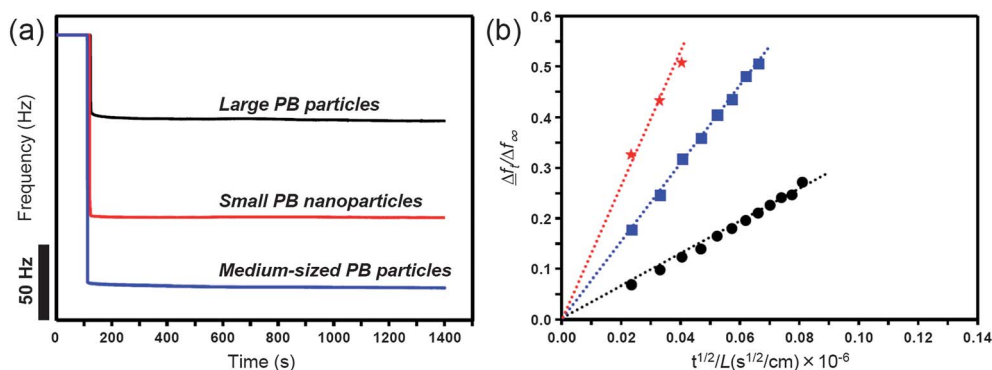


Fig. 13 (a) Time-dependent frequency shifts of QCM electrodes coated with PB particles caused upon adsorption uptake of Cs⁺ ions. (b) $\Delta f/\Delta f_\infty$ as a function of $t^{1/2}/L$ for the adsorption uptake of Cs⁺ ions to PB particles coating on the QCM electrodes. Red, blue, and black lines indicate small PB nanoparticles shown in Fig. 2c, medium-sized PB particles shown in Fig. 2e, and large PB particles shown in Fig. 7l, respectively.

data were analyzed using Fick's second equation and the graph of $\Delta f_i/\Delta f_\infty$ versus $t^{1/2}/L$ was constructed (Fig. 13b). The gradient for each sample was 12.6 (for small PB particles), 7.68 (for medium-sized PB particles), and 3.66 (for large PB particles), respectively. The diffusion coefficient (D) for small PB nanoparticles ($31.1 \times 10^{-12} \text{ cm}^2 \text{ s}^{-1}$) was much higher than those for the medium-sized PB particles ($11.6 \times 10^{-12} \text{ cm}^2 \text{ s}^{-1}$) and large PB particles ($2.63 \times 10^{-12} \text{ cm}^2 \text{ s}^{-1}$). From this result, it was proved that, in the case of small PB nanoparticles, the adsorption uptake of Cs^+ ions is very fast and Cs^+ ions freely diffused inside the PB pores. However, the diffusion of Cs^+ ions into the medium-sized PB particles was slower, although the adsorption capacity was much higher.

4. Conclusion

We developed an easy way to control the size and morphology of PB particles. By systematically tuning the pH value of the solution, the concentration of $\text{K}_3[\text{Fe}(\text{CN})_6]$, and the addition of PVP, three different kinds of PB particles were obtained: small, medium, and large. The pH value of solution, the concentration of $\text{K}_3[\text{Fe}(\text{CN})_6]$, and the amount of PVP worked together to control the size and morphology of products via a multiple-step mechanism. These different kinds of PB particles showed distinct N_2 gas adsorption-desorption behaviors. The small and medium-sized particles exhibited much higher BET surface areas than the large ones, probably due to the hindrance of removing water molecules during degassing caused by the larger size of the particles and dense crystal structure. In addition, the medium-sized particles showed the highest porosity. Our results proved that PB, as a kind of coordination polymer, crystallized in a non-classical manner, which was different from that of many other inorganic crystals but similar to that of biominerals.¹⁵ A new technique to improve the performance of PBs was established by not only reducing the size but also preparing mesocrystals, which was one of the main goals of scientists working in the field of non-classical crystallization. We believe that the non-classical crystallization mechanism can dominate the syntheses of other coordination polymers, enrich their morphologies and structures, and improve their performance.

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